

CSE340: Computer Architecture

Assignment -3

Deadline: 26th April, 11:59 PM

Assignment must be handwritten. Scan and upload PDF in provided Google form

Question - 1:

Convert the following IEEE-754 number into a decimal number where 6 bits are allocated for the exponent field in the representation.

0x ABB9609

Question - 2:

Perform the arithmetic operations using the floating-point format. Consider 5 digits for the fraction part.

a. $64.2486 * 49.1832$

Question - 3:

Subtract -4.0210 from 28.4810 using IEEE-754 single-precision floating-point representation. Check if the result has overflow or underflow or none.

Note: Consider 5 digits while performing the decimal to binary conversion of the floating portion.

Question - 4:

Write the necessary RISC-V code to compare two floating point registers f1 and f2. If f1 and f2 are equal jump to a label called “**jumpEqual**” else jump to another label called “**jumpNotEqual**”

Note: both the numbers stored in f1 and f2 registers are double-precision floating point numbers.

UNGRADED Problems (Only for Practice, no need to submit with Assignment)

1. a. Why is a bias added to the actual exponent in the IEEE 754 floating-point representation, and how does this affect the encoding of both positive and negative exponents?

b. How does optimized multiplication improve efficiency and performance compared to traditional long multiplication, especially in terms of speed and computational complexity?

2.

	Instruction name
1	fadd.s
2	fsub.s
3	fmul.s
4	fdiv.s
5	fsqrt.s
6	feq.d
7	fle.d
8	flt.s
9	mul
10	mulh
11	mulhu
12	mulhsu
13	flw
14	fsd

You **must** answer the following questions for the explanation of each of the instructions mentioned above:

- (i) What do they do/ How they work?
 - (ii) Explain their syntax.
 - (iii) Write an example for each of the instructions.
3. “mulh” instruction can be used to check for 64-bit overflow - Explain how with necessary example.
4. Perform the arithmetic operations using the floating-point format
- a. $50.7869 + 79.83 - 29.58$
5. Find out the smallest positive number and smallest negative number for
- a. The single precision format
 - b. The double precision format
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- a. The single precision format
 - b. The double precision format